

A Factorised 3D U-Net in the Beamlet Frame for Proton Pencil-Beam Dose Calculation on CT

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Abstract. We describe a per-beamlet dose engine for DoseRAD2026 Task 3, CT to proton dose. One model serves both anatomical sites and every beam energy. Each beamlet is resampled into a cuboid aligned with its own ray, $384 \times 64 \times 24$ voxels at $(1, 1, 3)$ mm, anchored where it first reaches the body. Its three inputs are CT in Hounsfield units, the beamlet energy and an analytic Bragg depth-dose prior computed with the dataset’s HU-to-density curve. A five-level encoder-decoder of residual GroupNorm/SiLU blocks with anisotropic pooling (53.26 M parameters) feeds a factorised head that emits an integrated depth-dose curve multiplied by a depth-conditioned lateral softmax kernel, making the depth-dose curve an explicit output with exact zeros beyond the distal fall-off. The loss is a dose-weighted MSE plus $0.05 \times$ a beamlet-frame integrated-depth-dose distance. Training uses AdamW (peak 3×10^{-4} , weight decay 10^{-5}) under a one-cycle schedule, effective batch 40, bf16 mixed precision, 240 epochs from random initialisation (the base model), with no pre-training, external data or augmentation; the submitted model adds three warm-start epochs under a rendered-frame bias term. Labels are pre-compensated for resampling by 32 Landweber iterations. Post-processing renders predictions onto the CT grid, applies a filled body mask and zeroes dose below each beamlet’s cutoff. Architecture and hyperparameters were fixed on a 61/6/8 patient split and the epoch by validation beamlet MAE; the base model retrains on all 75 patients. Code and weights: <https://github.com/maximfyn/beamlet-frame-proton-dose>.

Keywords: Proton dose calculation · Beamlet-frame regression · Factorised 3D U-Net · DoseRAD2026.

1 Introduction

Radiation therapy is one of the principal curative treatments for cancer, and the accuracy of its dose calculation is foundational to treatment plan quality. Proton therapy makes that accuracy harder to reach. A proton deposits most of its energy at a finite depth, the Bragg peak, which spares organs beyond the target, but that depth follows the integrated stopping power along the ray. A

range error of a few millimetres therefore does not blur the dose, it moves it: the peak lands short of the target, or past it in tissue meant to be spared. Monte Carlo (MC) transport is the accuracy reference and the speed bottleneck, at hours per plan on a CPU and seconds on a dedicated GPU engine [3, 8]. Learned engines predict a single beamlet far faster [7, 5], but the dataset’s authors name accuracy and real-time latency *together* as the problem still open [8]: in online adaptive workflows [6], an engine re-run per fraction is judged on latency as much as on agreement.

The DoseRAD2026 challenge releases beam-level MC dose for paired CT and MRI across thoracic and abdominal cases [8]. Task 3 asks for the dose map of a single proton beamlet on CT, scoring masked beam MAE and integrated depth-dose (IDD) distance per beamlet, then plan MAE, gamma and a dose-volume histogram (DVH) score on the composed plan, alongside a fitted runtime that carries twice the weight of any single accuracy metric.

A beamlet — a single proton spot, one energy at one lateral position — is a depth-dose curve times a lateral spread, both defined relative to its own ray. Asked to predict it on the patient grid, a network spends capacity locating the beam before it can model deposition, and must do so again at every gantry angle; learned beamlet engines therefore work in a frame aligned with the ray [7, 5], and so do we. That frame’s axes are the physical axes, so the integrated depth-dose curve — one of the five scored quantities — is a plain sum over the two lateral axes. We make it a factor the network emits and trains against directly, rather than a projection read off a voxel-wise output that a voxel-wise loss cannot control. The frame also bounds the problem: the box is small and fixed, so a modest 3D U-Net [2] suffices and inference stays cheap. Its one drawback is the round trip in and back, whose sub-voxel misalignment exceeds a good model’s entire error budget; pre-compensating the labels removes it once, offline.

We report one model serving both anatomical sites and every beam energy, and an analysis of where it falls short. The DVH score is the output we could least diagnose: it reads the planning target volume (PTV) and the nearest organs at risk (OARs) through structure information the challenge does not distribute, so it cannot be computed locally at all. The correction we trained for it improved the returned score, but did not survive a re-run. And the challenge metric divides each beamlet’s error by that beamlet’s own peak, so it weighs the weakest beamlets most while our loss does the opposite; rewriting the loss to match it changed nothing. Training and inference code, both model weights and the configurations that produced them are public at <https://github.com/maximfyn/beamlet-frame-proton-dose>.

2 Methods

2.1 Data

We used the DoseRAD2026 proton training release for CT only: 75 patients (39 thoracic, 36 abdominal), each with a CT at (1, 1, 3) mm, ray geometry and

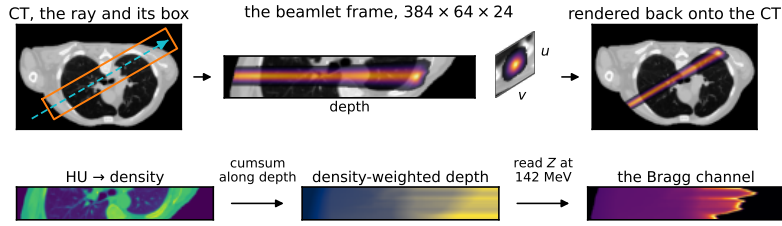


Fig. 1. Top: one real beamlet (thoracic, 300° gantry, 142 MeV), left to right: the CT slice with the ray and its box outline; the beamlet frame, CT in grey and ground-truth dose in colour, as the (depth, u) plane at $v = 0$ beside the (v, u) cross-section at the Bragg peak, drawn as the box’s end face with v receding; and that dose rendered back onto the CT grid, which is what is scored. **Bottom:** the third input channel, on the same beamlet. Its peak bends laterally because the depth-dose lookup is per voxel, not per depth.

energy for 1,080 beamlets, and their matching MC dose maps — 81,000 labelled beamlets. **No external patient data and no pre-trained weights of any kind were used**, for training, initialisation or as a feature extractor. The one external input is the public **Generic** machine table behind the Bragg prior, published long before the challenge.

The 75 patients were split once into a fixed, stratified registry of 61 train / 6 validation / 8 held-out test (29/32, 3/3 and 4/4 abdominal/thoracic). **This split is the subset on which all hyperparameter and architecture decisions were made:** every development run trained on the 61, selected on the 6 and scored on the 8. Holding it fixed makes those comparisons paired, which matters because patients differ far more than configurations do (Sect. 4). The submitted model is the one exception: its base retraining on all 75, so those 8 are training data for it, and every locally measured number here comes from a development run that never saw them.

Preprocessing is geometric and identical at training and inference time, up to the label; Fig. 1 is the data path. For each beamlet we build an orthonormal frame from the vector between its ray endpoints. Its first axis is depth along the ray, the second is lateral in the transverse plane, and the third runs along world z , the cranio-caudal axis, which is perpendicular to every beam in this dataset. In that frame we sample a $384 \times 64 \times 24$ box at $(1, 1, 3)$ mm, anchored where the box first reaches the body ($\text{HU} > -500$). Three channels are formed: CT clipped to $[-1000, 2000]$ HU and scaled, the beamlet’s energy — one scalar, normalised by 250 MeV and broadcast as a constant field over the box — and a Bragg depth-dose prior. The prior is built in three steps. HU is mapped through the ten-anchor HU-to-density curve released with the dataset, which is the calibration of the release’s own simulation, and integrated along the ray. The result is a depth coordinate related to water-equivalent depth, not that depth itself — density is not stopping power — and it tracked the real integrated depth dose better than the usual linear approximation, $1 + \text{HU}/1000$. At that coordinate we read the

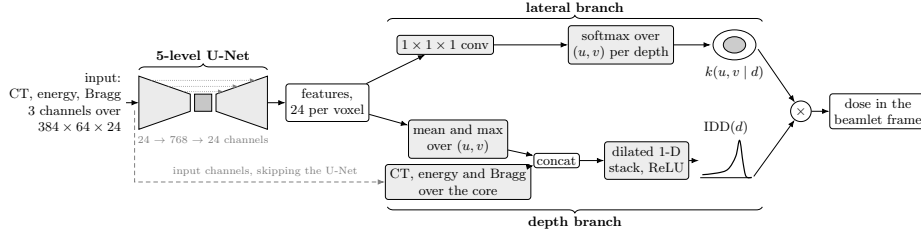


Fig. 2. The network. A five-level U-Net turns the three input channels into 24 features per voxel. Two branches read them: the depth branch emits $\text{IDD}(d)$, how much dose lies at each depth, and the lateral branch a kernel $k(u, v | d)$ that a softmax normalises to unit mass at every depth, its shape there. The dose is their product. The depth branch also reads the input channels pooled over the core, the central 16 of 64 voxels in u and 6 of 24 in v , skipping the U-Net and drawn as such.

depth-dose curve of the **Generic** proton machine distributed with pyRadPlan [1], the same machine the dataset paper names for the release’s beam parameters [8]. The prior is a feature only: nothing is regressed onto it.

Labels come from the released MC dose maps d on the CT grid. Two resamplings separate a beamlet-frame prediction from its score: the truth is sampled into the box by an operator S , and a prediction is rendered back onto the CT grid by an operator R , both trilinear. Their composition is not the identity, $RSd \neq d$, and the loss is systematic: it blurs the steepest gradients, which sit at the Bragg peak. Trained on the naively sampled label Sd , a network would learn a target whose render already misses the truth. On 800 beamlets of the 8 held-out patients, that label alone, rendered back, scores a beam MAE of 0.0098 — above the error of the submitted model (Table 2). We therefore solve instead for the non-negative beamlet-frame array whose render reproduces the truth, $\min_{x \geq 0} \|Rx - d\|^2$, by projected Landweber iteration [4]: $x_0 = Sd$ and $x_{k+1} = \max(0, x_k + \omega S(d - Rx_k))$, with $\omega = 0.8$ (not tuned) and S standing in for the adjoint of R . We call the result *pre-compensated* and use x_{32} as the label. The first iteration halves this floor, and 32 iterations lower it by $\sim 85\%$, to 0.0014. This is computed once, offline: training is then a plain regression in the beamlet frame, and inference applies R once to each prediction.

2.2 Model

The network is a factorised 3D U-Net with **53,259,786 parameters** (53.26 M) in 39 **Conv3d**, 5 transposed-convolution and 3 **Conv1d** layers, in PyTorch 2.9.1 with NumPy and SciPy for geometry and SimpleITK for I/O. All weights are trained from random initialisation, and no external or pre-trained weights enter at any point; the one departure is the depth branch’s output convolution, zero-initialised with a small positive bias so the depth curve starts flat and every ReLU is alive at step zero.

U-Net (Fig. 2). A five-level encoder–decoder over the three-channel cuboid. Each level is a residual block of two $3 \times 3 \times 3$ convolutions with GroupNorm (8 groups, not BatchNorm: at this crop the per-device batch is set by memory rather than chosen, and was halved when v was widened from 16 voxels to 24) and SiLU, plus a $1 \times 1 \times 1$ projection on the shortcut; widths double at each of the five downsamplings, from 24 to 768. Downsampling is a strided convolution and is **anisotropic**: strides (2, 2, 1), (2, 2, 1), (2, 2, 2), (2, 2, 2), (2, 2, 2) in (depth, lateral- u , lateral- v), protecting the 24-voxel v axis while depth is reduced $32\times$. The decoder mirrors it with transposed convolutions and concatenating skips at every resolution.

Head (Fig. 2). The decoder output becomes dose as $\text{IDD}(d) \times k(u, v \mid d)$. The lateral branch produces k with a $1 \times 1 \times 1$ convolution to one logit per voxel followed by a softmax over (u, v) at each depth, so it carries unit mass and is free in shape at each depth. The depth branch takes the U-Net’s features mean- and max-pooled over (u, v) , concatenated with the input channels pooled over the central core (16 of 64 voxels in u , 6 of 24 in v), and runs a three-layer dilated 1-D stack (kernel 5, dilations 1/2/4, 64 features) with a ReLU output.

Because k sums to one at every depth, $\sum_{u,v}$ of the product is identically $\text{IDD}(d)$: the factorisation re-parameterises the output rather than restricting it, and puts the scored curve where the loss can reach it. It assumes neither separability nor a Gaussian: lateral sigma grows from ~ 3.9 to ~ 6.9 mm between entry and peak, and a single Gaussian leaves a 12% L_1 residual. ReLU rather than softplus is deliberate: a zero depth slice zeroes a whole slab, most of the box past the distal fall-off.

Post-processing. The network predicts in the beamlet frame, so each prediction is rendered back onto the CT grid, on that plan’s own spacing, and masked with a body contour that is filled slice by slice, so that air inside the patient is kept rather than cut away: the labels carry dose there. Everything below that beamlet’s served minimum cutoff is then zeroed, as the challenge requires of every dose map.

2.3 Training

For one beamlet, y is its pre-compensated label and $\hat{y}$ the network’s prediction, each the full 3D dose array over the $384 \times 64 \times 24$ box in the beamlet frame. The objective is $\mathcal{L} = \mathbb{E}[w(\hat{y} - y)^2] + 0.05 \text{RMS}_d[(\text{IDD}(\hat{y}) - \text{IDD}(y))/\max_d \text{IDD}(y)]$, where the expectation runs over the voxels of every beamlet in a batch, $\text{IDD}(\cdot)$ sums an array over its lateral axes (u, v) to a curve over depth d , and $w = \max(y/\max y, 0.05)$ is normalised to unit mean per beamlet, so voxels carrying dose dominate while the empty region still contributes enough that predictions do not bleed outside the beam. The second term is the IDD metric’s own arithmetic in the beamlet frame, a sum over (u, v) and so exactly differentiable.

Hyperparameters and architecture were chosen by single-axis sweeps on the 61/6 split — over the input channels, the HU conversion behind the prior, the head, the number of levels, the IDD loss weight, the box’s lateral extent, the Landweber count and the epoch budget — each variant scored on the 8 held-out

Table 1. Configuration of the base model: one model for both anatomical sites and all beam energies. One-cycle warms up over the first 10% of steps. The submitted model is this checkpoint plus the three warm-start epochs described in the text.

Input	$384 \times 64 \times 24$ at (1, 1, 3) mm; CT (HU), energy, Bragg prior
Architecture	factorised 3D U-Net, 5 levels, width 24, 53.26 M params
Loss	dose-weighted MSE + $0.05 \times$ beamlet-frame IDD distance
Optimiser / LR	AdamW, weight decay 10^{-5} ; one-cycle to 3×10^{-4}
Batch / epochs	effective 40 (20×2 GPUs); 240, best epoch 229
Precision / aug.	bf16 autocast; none
Init. / data	random, no pre-training; all 75 patients (81,000 beamlets)

patients, with choices for the submitted model also informed by preliminary-phase feedback; Table 1 gives the final settings. **No augmentation and no ensembling were used in the submitted model.** Both were measured. A voxel-mean ensemble of three checkpoints was accurate but is $N \times$ the forward path on the term that dominates the fitted runtime, which carries twice the weight of any accuracy metric; augmentation is taken up in Sect. 4. The base checkpoint is the epoch minimising validation beamlet MAE in the beamlet frame, weighted to the test set’s published 14/26 thoracic/abdominal mix [8], since our splits are even; the checkpoint records the selector. That validation set is held out for every development run; for the base of the submitted model, which trains on all 75, it is not, so its epoch was chosen on patients it had seen.

The submitted model is that checkpoint plus three warm-start epochs at 3×10^{-5} on the 61 development patients — its base had trained on all 75, so nothing is truly held out — under one added loss term at weight 32: the squared dose-weighted moments, of orders 1 to 3, of each beamlet’s peak-normalised signed residual after rendering onto the CT grid. The last of the three epochs is submitted. What it was for, and what it did, are in Sect. 3.

2.4 Evaluation

Level 1 is masked beam MAE and IDD curve distance per dose map, scored locally by an implementation held to exact equality with the challenge’s released evaluation code. Level 2 (plan MAE over three dose strata, local 1%/1 mm gamma, and a DVH score over PTV $D_{98\%}/V_{95\%}$ and the three nearest OARs) needs beamlet weights and structure masks that are not distributed, so we report only what the organisers returned. Runtime is reported as the challenge computes it: a non-negative least-squares fit of $T = t_{\text{fix}} + N_{\text{im}}t_{\text{im}} + N_{\text{map}}t_{\text{map}}$ read at one image and 500 dose maps.

Two conventions govern the numbers below. The $\pm$ on any challenge metric is a population standard deviation over plans, a between-patient spread and not a confidence interval; the preliminary phase serves six patients and the final test forty, so a mean is known to roughly $\sigma/\sqrt{6}$ and $\sigma/\sqrt{40}$. And comparisons are judged against a measured rerun floor: rerunning a configuration, with a fresh

Table 2. Organisers’ evaluation of the submitted model on both test partitions; $\pm$ is a population standard deviation over plans, not a confidence interval.

	final test ($n = 40$)	preliminary ($n = 6$)
beam MAE $\downarrow$	0.0065 ± 0.0017	0.0087 ± 0.0020
IDD distance $\downarrow$	0.0030 ± 0.0014	0.0033 ± 0.0016
plan MAE $\downarrow$	0.0054 ± 0.0016	0.0073 ± 0.0023
gamma $\uparrow$	97.65 ± 2.31	96.74 ± 3.12
DVH score $\downarrow$	0.376 ± 0.303	0.768 ± 0.338
runtime	17.64 s	16.83 s

random initialisation (no seed is fixed), moves beam MAE by 1.2% and IDD by 0.4%, and a pure replicate can come out ahead on all eight held-out patients, so a unanimous win does not make a difference real. No explainability analysis was performed.

3 Results

3.1 Challenge partitions

Table 2 gives the organisers’ returned metrics for the submitted model on both partitions. The final test holds 40 patients, 26 abdominal and 14 thoracic; the preliminary test six, three of each.

The submitted model cannot be scored against held-out patients, since its base trained on all 75, so the numbers below come from the development model (not released) those decisions were made on: 120 epochs, on the narrower box the sweeps started from (16 rather than 24 voxels in v). Those 8 patients are also where every configuration was chosen (Sect. 2.1), so these are selection estimates, optimistic for the configuration chosen. It reads beam MAE $0.00621 / 0.00734 / 0.00756$ on the 61 training, 6 validation and 8 test patients: validation runs 2.9% ahead of test because every run selected its checkpoint there, and training 20.23% ahead of the two pooled, a generalisation gap taken up in Sect. 4.

3.2 Runtime

The submitted container is scored at **17.64 s** for one image and 500 dose maps on the final test (16.83 s on the preliminary), an amortised 35 ms per dose map. That is per beamlet, not per plan: a clinical plan carries thousands of them. Its allocation and scheduling optimisations are voxel-neutral: with every one on and then off, the container writes byte-identical volumes. Replaying the platform’s job shapes on an AWS **g5.2xlarge** instance (NVIDIA A10G) matching the evaluation hardware, inference is 97% of the wall clock and the forward pass $\sim 78\%$ of that. At 2,519 GFLOP per batch of 8 the network is compute-bound (211 FLOP/byte against the A10G’s ~ 58 balance point), and its convolutions already run on TF32 tensor cores: disabling TF32 doubles the forward time.

Reduced precision goes further — against the eager forward, fp16 autocast takes $0.83\times$ the time at batch 8 with cuDNN autotuning ($26.1 \rightarrow 21.7$ ms per beamlet), with its effect on dose untested here — so runtime is available from precision as well as from fewer voxels or channels; ahead-of-time compilation, already shipped, takes $0.855\times$. Training is a one-off cost of 49.1 h for the base model and 0.6 h for the three fine-tuning epochs, each on two NVIDIA H200 GPUs.

3.3 Outputs with poor performance

DVH score: the output we could least diagnose. It needs beamlet weights and structure masks the challenge does not distribute, so unlike beam MAE and IDD it cannot be computed locally at all. It is also a threshold statistic on the PTV and three organs at risk rather than a volume average, so it answers to the high-dose region in a way the averaged metrics do not. Both are properties of the metric, not of how long the model trained.

The DVH correction: it did not replicate. The added term reaches what the beamlet frame cannot, since the DVH score is read on the composed plan, in the CT frame. It is a patch on a finished model rather than a design choice: retraining from scratch under it would have cost days of GPU time the window did not leave, so Table 1 remains the recipe the network was built by. On the preliminary partition the returned score moved $0.813 \rightarrow 0.768$, -5.6% . Re-running that recipe from the same base checkpoint, differing only in run-to-run non-determinism, recovers almost none of it. On six patients a move of that size is what a rerun alone produces, so we report it and claim no mechanism.

A better term: the one we used is stronger than the metric needs. What the DVH score requires is that a patient’s rendered residuals sum to about zero across the whole plan; what the term penalises is each beamlet’s own residual moments. Those are different conditions, because a plan is a sum of beamlets: one biased high can be cancelled by one biased low, leaving the composed dose unbiased though neither beamlet is. Squaring per beamlet scores both as errors, treating “every beamlet high by 0.09%” and “half high, half low” alike though the first composes to a 0.09% plan offset and the second to nothing. Pooling the signed residual within a patient before squaring targets the composed quantity directly.

Bragg-peak bias: real on the checkpoint we could measure it on. Over 8,640 held-out beamlets, an earlier checkpoint on the narrower box under-predicts the peak at low energy and crosses over at high: median predicted/ground-truth 0.978 below 80 MeV, rising monotonically to 1.012 above 199 MeV. The label pipeline is not responsible, since rendering a pre-compensated ground truth reproduces its own peak to within 0.27% in every box and iteration setting tested, so a peak ratio measured on a checkpoint is a property of the network. The submitted model has no held-out cohort among the 75, so whether it carries the same bias is unmeasured here.

Beam MAE: the metric scores each beamlet relative to its own peak, so it weights weak beamlets most, while our dose-weighted loss weights strong ones most. Over the 8,640 held-out beamlets our absolute error is flat across quartiles

of true peak height (7.9, 8.6, 8.1, 8.0×10^{-6}) while the metric falls from 0.01070 to 0.00541, a log-log slope of -0.954 against that height. Matching the two did not help: a loss that divides the residual by each beamlet’s own peak, as beam MAE does, left beam MAE unchanged in a paired run (+0.8%, better on 4 of 8 patients).

4 Discussion

The first limitation is statistical. Between-patient spread in beam MAE is 13.5% of the cohort mean on our 8 held-out patients, against differences between configurations of 1–7%, so a difference read from a single submission is small relative to the variation between patients. Comparisons here are therefore paired on a fixed split and judged against a measured rerun floor.

Two differences between the training and test distributions affect generalisability. **Site mix:** our held-out partition is 4/4 thoracic/abdominal and the preliminary phase 3/3, whereas the final test pool is 65% abdominal, so both over-weight thoracic. **Plan structure:** the released training plans are a synthetic grid of 36 gantry angles drawing on one of two energy sets, one per anatomical site; the preliminary phase served clinical intensity-modulated proton therapy plans of three gantry angles, with energy ranges that differ from patient to patient. The model is per-beamlet, so plan structure never reaches it, but the induced energy distribution does: median beamlet energy on the served plans is 77.9 MeV abdominal against 138.6 MeV in training, stopping at 147.1 MeV where training reaches 200.8. They lie *inside* training’s support, so this is interpolation, not extrapolation.

Generalisation: the development model reads +20.23% from its training patients to the held-out ones, and pooling all 75 would have hidden it — that reads 13.7% “better” on 81% memorised weight, so a model trained on more of them comes out ahead for that reason alone. Every comparison here is held-out only. The same gap read on IDD is +7.49%, 0.9 standard deviations and so within that metric’s own noise. The gap can be reduced: an arm with flip augmentation, weight decay 10^{-3} and an exponential moving average of the weights halves it to +10.44% at no inference cost, and was the only intervention we tried that moved it. That arm was built to measure the gap rather than to be submitted — measuring one needs held-out patients, so it trained on the 61, and for 120 epochs, where the submitted model’s base trained on all 75 for 240 — so the three regularisers stay confounded with each other and with the box.

Training the base model on all 75 patients bought 23% more data for a hidden test set at the price of local scoring: the two metrics that can be computed locally have no held-out patients left to measure it on.

Two experiments remain unrun, neither of which fitted the available time. A matched regularisation arm, trained for the same epochs on the same patients as the submitted model, would establish whether the reduced gap yields a better score or only a smaller gap. A model trained from scratch under the rendered-frame term, rather than fine-tuned with it for three epochs, would establish

whether the DVH score is reachable from a beamlet-wise objective at all. On the evidence reported here, either could outperform the submitted model.

5 Author contributions

Maxim Braunsburger is the sole author and contributed all applicable CRediT roles.

6 Other information

No funding was received and there were no external collaborators. GPU training ran on the Interdisciplinary Center for Scientific Computing (IWR) cluster, Heidelberg University, with CPU jobs at the German Cancer Research Centre (DKFZ); runtime was measured on an AWS **g5.2xlarge** instance. Niklas Wahl, one of the challenge organisers and a co-author of the dataset paper [8], leads the author’s group at DKFZ and had no role in this work. While preparing this work the author used AI assistants for software development and to draft and revise the manuscript, and reviewed and edited all content, taking full responsibility for it.

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